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Vulnerability Assessment Report

Learning vulnerabilty analysis, risk classification through FutureInterns Cybersecrity internship

Steps in vulnerability analysis

Asset identification

Involves scpe definition hrough categorizing resources i.e, hardware, software, network and cloud components

Vulnerability scanning

Involves use of automated tools such as scanners, e.g Nmap. qualys for different IT evironments.

Analysis & Prioritization

Asssessing the risk level of each vulnerability based on potential impact and exploitability

Remediation & Reporting

Creating a detailed report outlining recommendations to fix, patch, miigate and curb the ipotential areas of weaknesses.

Common web vulnerabilities

Deception

Involves social engineering tactics to manipulate individuals into providing information or divulding sensitive data to unauthorzed persons

Cyber attacks

Use of malicious software or malware to carry attacks. Malware is any code used to steal data, bypass access or compromise systems

Wireless and mobile device attack

Wirelessly compromising sysyem or data through bluesnarfing, rogue AP, RF jamming etc.

Application attacks

use various techniques to attack application, i.e XXS, code ijection, buffer overflow etc

Risk classification

Low level attacks

Limited impacts, often aimed at low hanging Fruits. These includes : generic spam/phising, Adware, port scanning

Medium level attacks

These are targeted and can cause operational disruption, financial loss or theft of non-critical data. These are: DDOS attacks, MITM,Brute force, watering hole attack

High level attacks

Sophisticated targeted attacks that result in critical damage, mass data breaches or loss of control over sysytems. Examples, APTs, Zero day exploits, ransonware etc.

Evaluation of cyberthreats on businesses

Financial loss
Direct costs from ransonware payments, legal fees and opeartional downtime which can affect overall revenue.

Reputational damage

Loss of customer trust, reduced stakeholder can cause long years to recover.

Operational disruption

Cyberattacks e.g DDOs can freze critical business services and halt production

Theft of intellectuual property

Compromise of trade secrets, research and confidential data can lead to competitive disadvantages.

Securing networks

Employee awareness

Internal threats can be curbed through proper employee awareness by educating against phishing scams and engineering attempts.

Access control and authentication

Implementing MFA for all users, enforce strict password policies and apply the principle of least priviledge to restrict user access to only necessary data.

Response planning

Establishing a robust incident response plan to ensure minimal downtime and fas recovery from attacks.

Nerwork segmentation

Dividing the network into smaller segments to limit the impact of a breach and restrict lateral movement by atckers.